TAIWAN
Regional Community

The Green Tokens:
From Green Computing to Sustainably Intelligent

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ISO/IEC 21031:2024
ISO 30134
ISO 14040/14044



Green Tokens



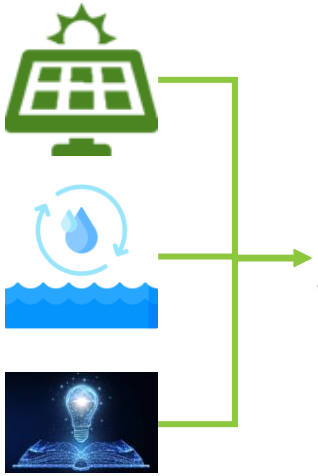
Is AI for people, or are people for AI?

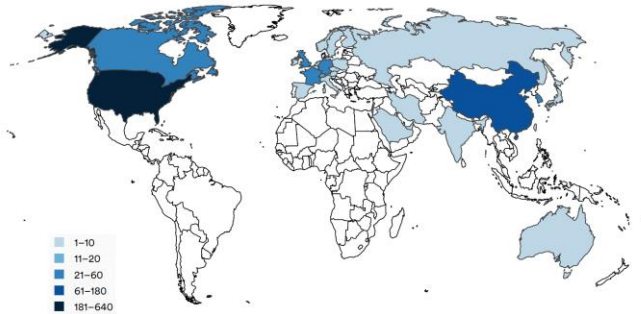
- I think there has been a huge debate since AI was created**: Is AI for people, or are people for AI?**
- I do not want to conclude this debate, but I try to use the term **'green tokens' to remind people that the key is in our hands** and we have the opportunity (option) to make the right choice.
- **Green** means the **universal values for good**. What does 'sustainably intelligent' mean? I think it means that **the next generation can enjoy the benefits of AI just like we do**.



What is Green Tokens?

- **Green tokens** is a **disclosure mechanism** for digital service applications ex. AI.
- Green tokens is a **tracking** for the AI service which let the people know **when, how, and where**, the provided service was generated.
- Green tokens is also a mechanism which the **universal concept of good** can be embedded into **AI services through option**.

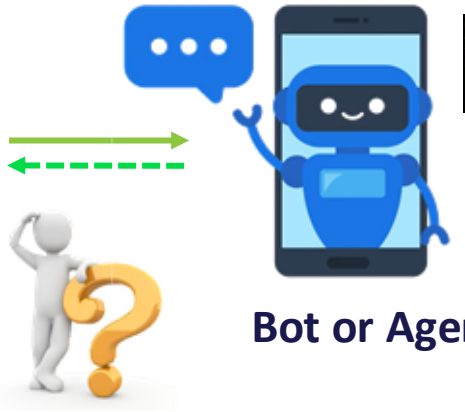




Number of notable AI models by geographic area, 2003–25 (sum)

Source: Epoch AI, 2026 | Chart: 2026 AI Index report





Bot or Agent

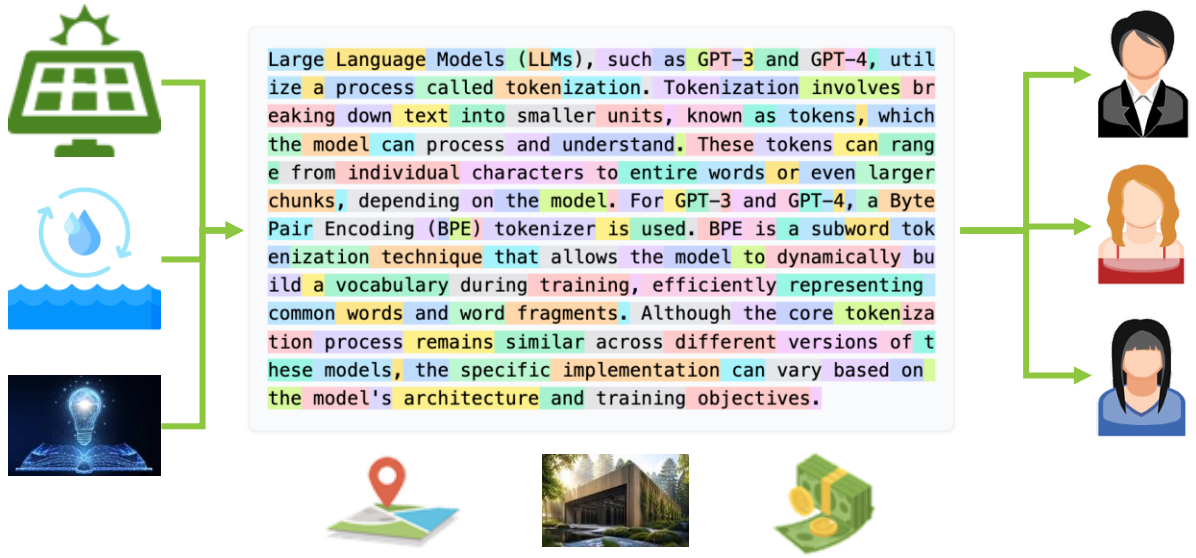
Tokens





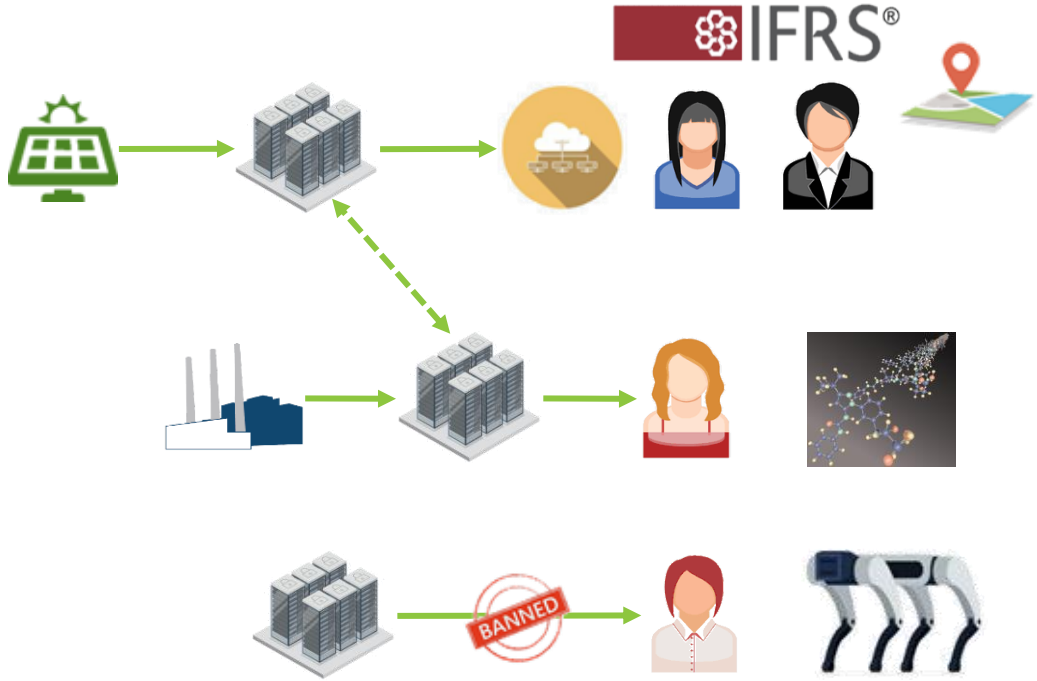
Why the matter of Green Tokens for End-users?

- Green tokens helped end-users to **recognize the attributes of each AI token which they used** and understand the resources consumption and impact to the environment during specific period of time when they enjoy the powerful efficiency of AI.
- Green tokens can help the end user to **see their behavior impact to energy and water resources, or environmental impact.**
- Green tokens might also help the user to **see their contribute of carbon reduction when they changed their behavior**.



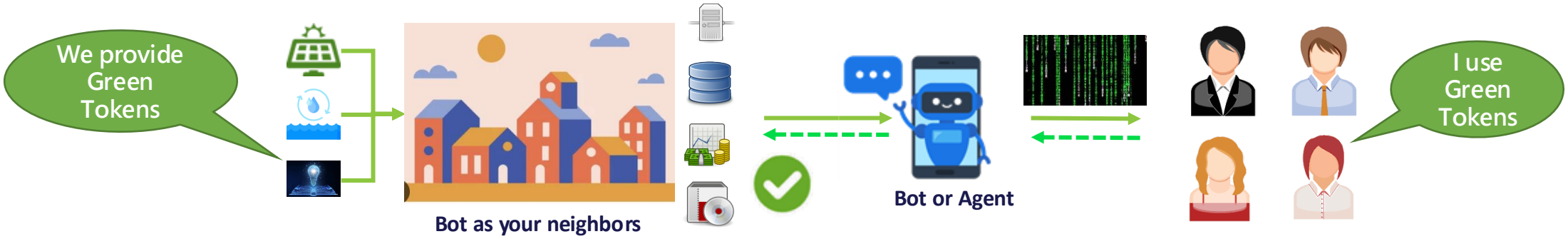
Why the matter of Green Tokens for AI service Providers?

- Green Tokens can help the AI service provider to really understand what exactly the **potential cost & benefit** for different kinds of user and application **when users are capable to make choice**.
- Green Tokens might be the **social perception** of AI service provider and help them to **design or differentiate products for user with different preference, and** to help service provider to allocate their resources to serve the right customer in a **more 'beneficial' way**.



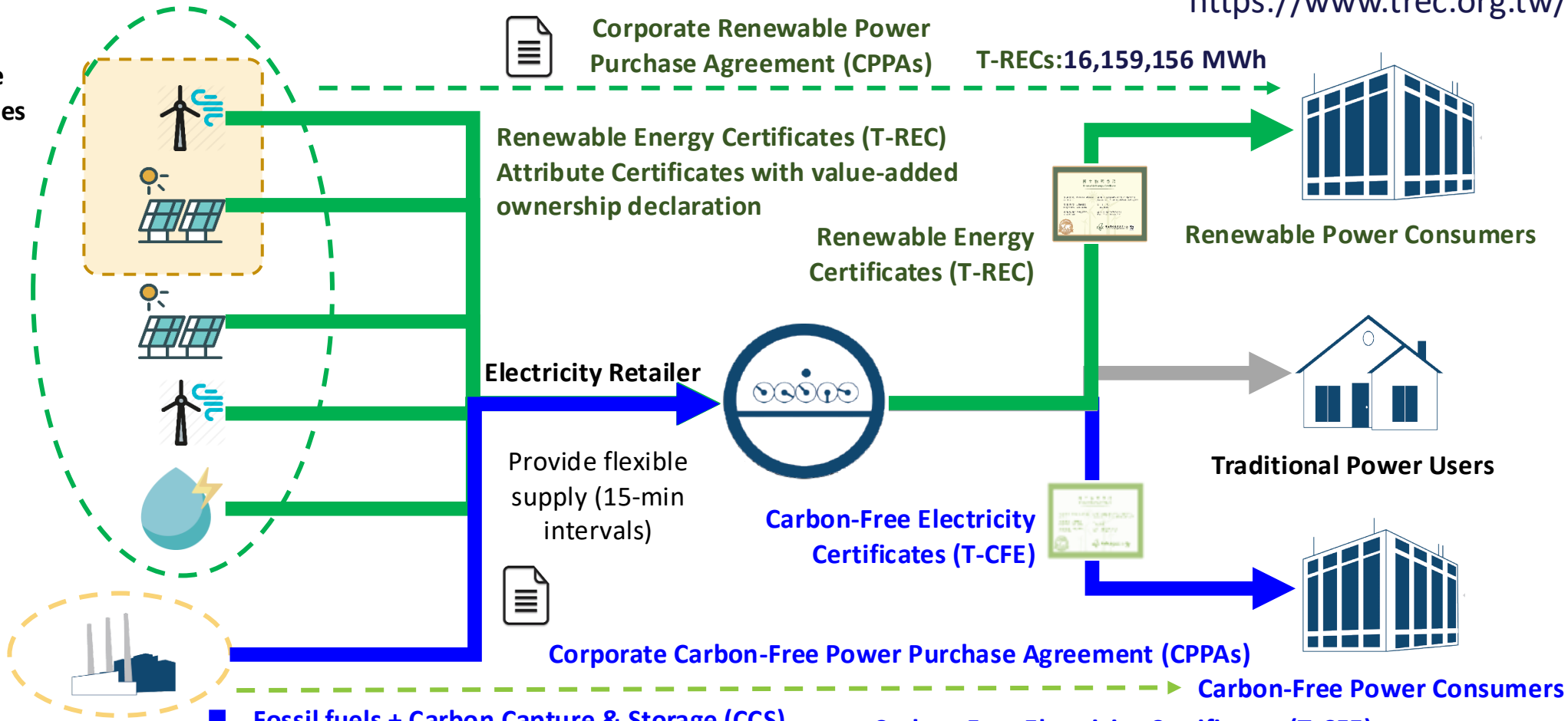
Why Green Tokens is essential?

- **There's no turning back for development of the AI.** But human (you and me) should be the guy to indicate which way to go.
- We hope not only the AI provider make the decision of direction. We should also encourage the public to join in this process as a beacon, so that the end-user will not only contribute their knowledge, **but also their consciousness about universal concept of good, responsibilities, integrity etc.**
- Maybe more people will not be skeptical about the AI and become more favor of AI, **when Green Tokens become the critical of AI**.



Case Study 1: New 24/7 Carbon-Free Electricity Economy

https://www.trec.org.tw/



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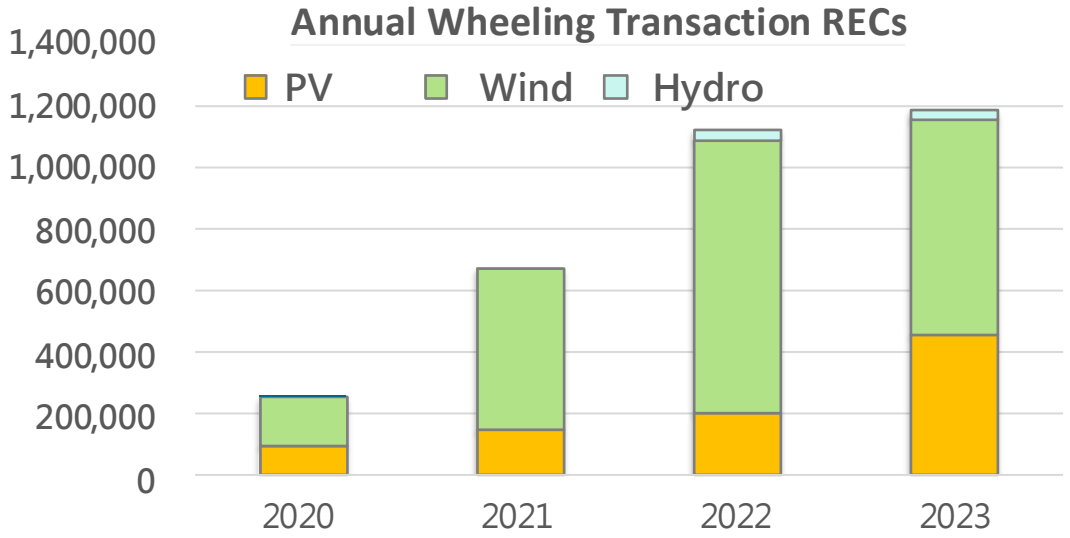
T-RECs Dashboard

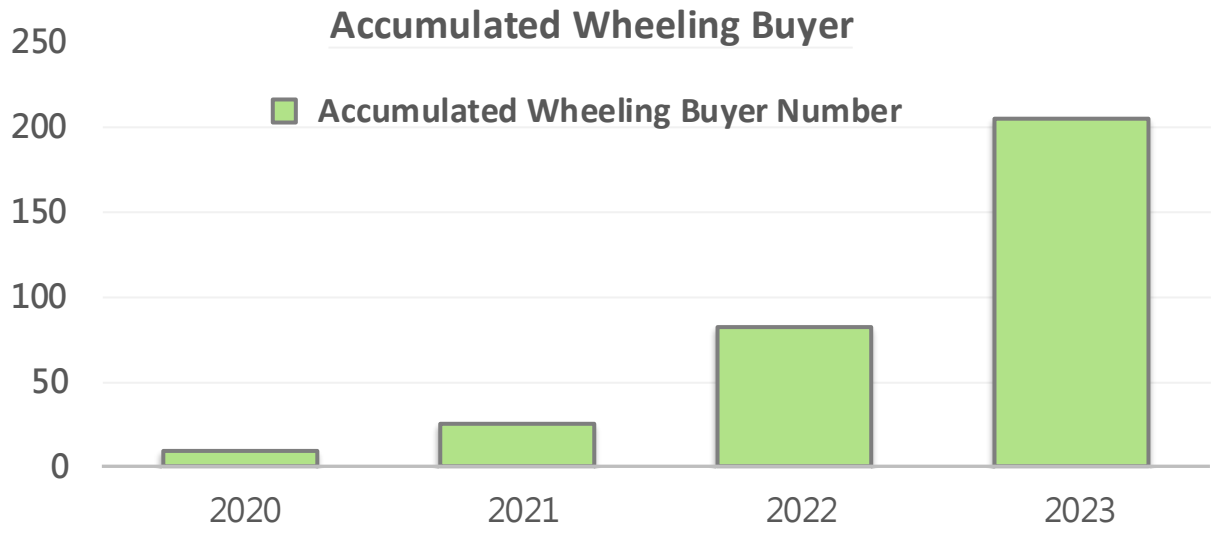
Since May 2020, 3,519,211 RECs have been traded & transferred (3.51 billion kWh, 1 REC represents 1MWh)

Annual Wheeling Transaction RECs					
year	PV	Wind	Hydro	Total	Increase (%)
2020	93,639	159,657	0	253,296	-
2021	146,085	525,757	0	671,842	165
2022	200,520	890,455	34,129	1,125,104	67
2023	455,120	702,055	31,262	1,283,242	-
Total	975,755	2,382,571	65,391	3,519,211	

Until September 2023, 205 Buyers has sourced renewable electricity by CPPA, Enterprise 75.1%, SME 20.0%, NPO/School 3.4%, Person 1.5%

Category	Number of Buys	Ratio (%)
Enterprise	154	75.1
SME	41	20.0
NPO / School	7	3.4
Person	3	1.5
Total	205	100%

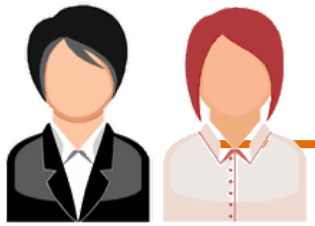




Case Study 2: Green Mileage support transportation sector towards net zero emission

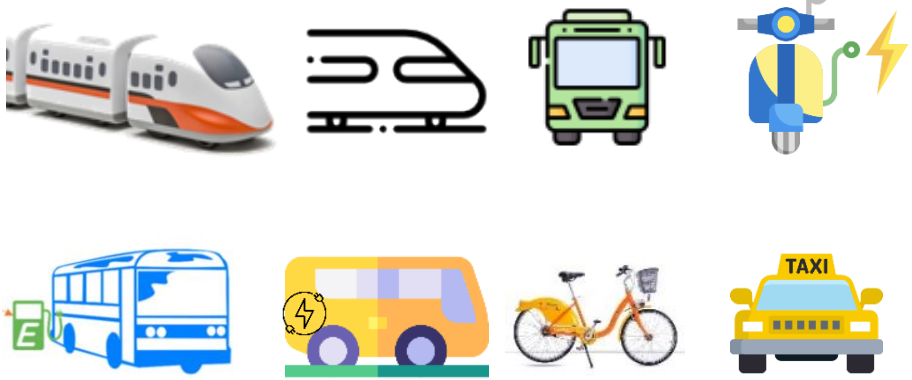
To promote of sustainable transportation mode choice , and promote the development of green transportation

Public & employees



Purchase Touring Package

Seamless and integrated low-carbon transportation services



Provide high-accuracy commuting information

- Transportation service users
- Vehicle type
- Green mileage/ Net Zero mileage

Government and Company encourages public and employees to adopt Green Mobility





Government & Employer provides subsidies to employees to adopt Green Mobility Service

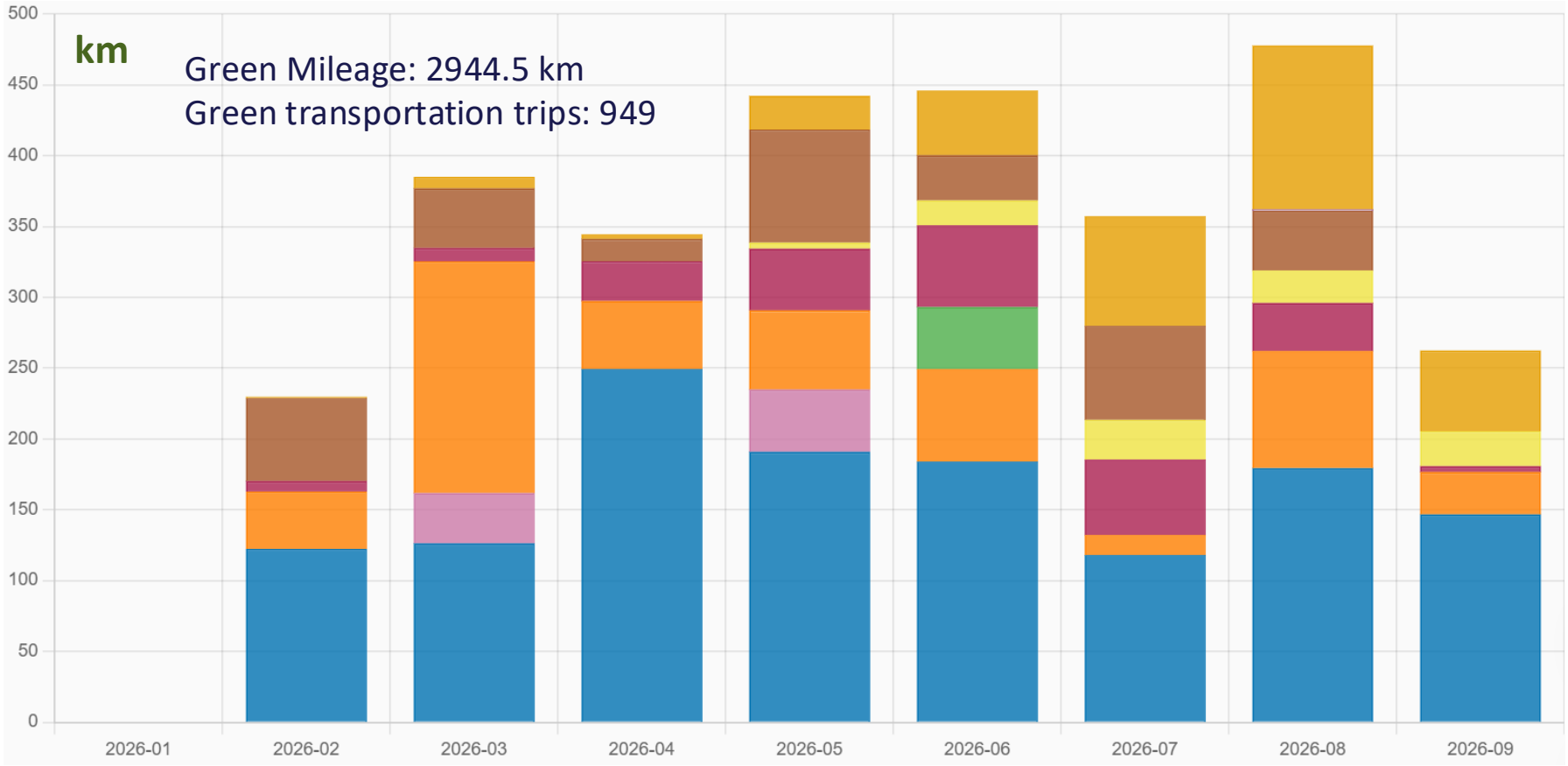
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Green Mileage Dashboard

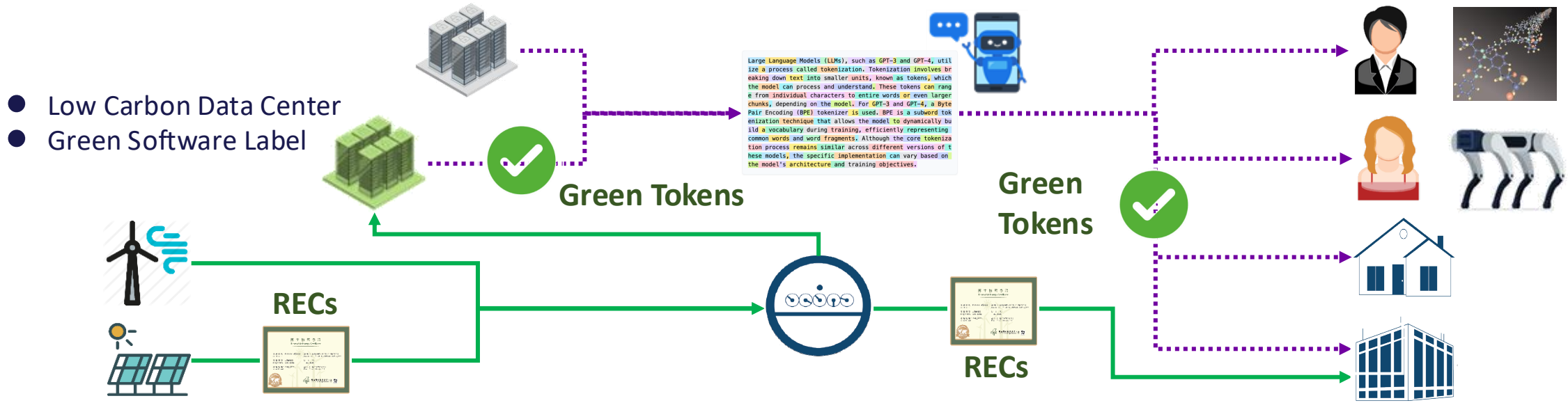
I am an environmentally friendly green Commuter.

- City Bus
- Intercity bus
- Metro
- Taiwan Rail
- City Bike
- Private Bike
- High-speed Rail
- Sharing E-Scooter
- Private E-Scooter
- Sharing EV
- Private EV
- Walk
- Taxi



Call for Case Study 3: Green Tokens Economy

- **It has been over a century since Thomas Edison pioneered electricity** here in the United States. Since then, the development of power generation, grid, metering, regulations, banking, markets, and REC systems has occurred step by step, gradually evolving into the sustainable power economy we see today.
- **Again, Generative AI was invented here in the US, but it is evolving ten times faster than electricity did.** Because of this, we can expect **parallel rather than sequential development**.
- Let's **create the Green Tokens mechanism and accelerate the evolution of sustainable AI together**.



Taiwan Voluntary Green Software Label Proof of Concept

- Developing Green Software Industry
- Green Software Talent Cultivation



Software Practitioners

- Green Software Labeling Mechanism
- Software Carbon Intensity Standard and Verification Mechanism
- Promoting Green Software Labeling and Green Token



Customized Digital Services

Green Software design capability

Customized | Product-Oriented | Environment Friendly



Standardized Cloud Services

Equal function unit environment friendly services

Standardized | Service-Oriented | Environment Friendly

- Enterprise Green Software and Applications Awareness
- Encourage Enterprise adapt Green Software and Applications Service

Consumer Awareness (Individual/ Enterprises)





- Select Environment Friendly Software
- Select Environment Friendly Operation Application Services

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Example: AI Green Tokens Dashboard

Green Tokens can be the foundation of following metrics of specific period time or per token:

1. **Location: community, domestic, abroad**



2. **Data center: green data center, normal data center**



3. **Electricity: consumption, renewable, low carbon, general**



4. **Emission: total emission**



5. **Load Immediacy : Load-shift computing, real-time computing**





Call to Action (1/2)

- **AI providers can:**
  - **Improve the transparency for AI using behavior of end-user.** Provide not only the throughput limitation, but also how many energy, resource consumption and impact happened undertaken by their AI behavior.
  - Provide end-user their contribution of AI using behavior changing, ex. scheduling AI workloads during periods of abundant low-carbon energy.
  - Differentiate the product for different user with different preference.

Call to Action (2/2)



- **AI end-user can:**
  - **Choose the right AI model to deal with the right thing.**
  - **Ask your** AI providers to provide the **green tokens mechanism**.
  - **Pay attention to how many green tokens you got and realize their impact to the resource or environment.**
- **Green Tokens interested Parties can:**
  - **Setup a workstream in OCP to develop the concept** of **Green Tokens mechanism and implement it together.**



Taiwan is developing this mechanism and invites community input toward future pilots.
Please contact: yenhaw@msn.com

Let’s create the Green Tokens mechanism and accelerate the evolution of sustainable AI together.

Thank You!

